

Foundations of Artificial Intelligence

M.Sc. Mathematics with Artificial Intelligence & Data Science

UNIT I • LECTURE 02

AI in Daily Life, Intelligence, Mathematics & Security

“How many of you used Google Maps today? How many unlocked your phone with your face?”

Let them raise hands. Do not explain anything yet. Let them see the numbers themselves.

“AI is not a robot standing in a corner. AI is what woke you up, showed you the road, paid your bill and chose your videos. It was working all morning. You just did not see it.”

6:00 AM	7:30 AM	8:15 AM	8:45 AM	All day
Your phone unlocks by looking at your face.	Maps picks the fastest road to college.	You pay by UPI and the bank checks it is really you.	YouTube shows videos it thinks you will like.	Spam is removed, feeds are ordered, prices are guessed.

AI Is Everywhere in Daily Life 8–18 min

Travel and movement - Google Maps — finds the shortest road - Uber / Ola — guesses how many people want a ride - Swiggy — plans the delivery route	Watching and shopping - Netflix, Spotify — suggest what to watch or hear - Amazon, Flipkart — suggest what to buy - Instagram — decides what appears first	Talking and working - ChatGPT — writes and answers - Google Translate — changes the language - Gmail — removes spam
Safety and health - Face Unlock — recognises your face - UPI, banking — catches fraud - Hospitals — helps find disease early	Class activity (3 minutes) “Where else did you use AI today?” Write every answer on the board. Then for each one ask a second question: “What is this app trying to guess?” This turns a list of app names into a list of real problems.	

Mini Case Study — The Day AI Stopped

Imagine this: tomorrow morning, every AI system in the world stops working. What happens?

Banking — payments stop, loans cannot be checked.

Transport — traffic jams, routes fail.

Hospitals — scan reading slows down badly.

Communication — spam fills every inbox.

What Is Intelligence?

“Who is an intelligent person?” Most students will describe someone who adjusts to new situations, not someone who remembers a lot.

Simple definition: intelligence is not only knowing things. It is the ability to learn from what happens, think properly, and handle a new situation you have never seen before.

1 · Observe	2 · Learn	3 · Reason	4 · Decide	5 · Solve & Adapt
You notice what is happening.	You understand the pattern.	You think it through.	You choose an action.	You handle it, and change for next time.

“Notice that step 5 goes back to step 1. Whatever you solve today becomes something you observe tomorrow. Intelligence is a circle, not a straight line. It never finishes.”

Machine learning uses exactly the same circle, only with different names: collect data → train the model → make predictions → check the results → train again.

What Is Artificial Intelligence? 28–38 min

Word	Meaning in simple language
Artificial	Made by human beings.
Intelligence	The power to think, learn and decide.
So, AI is...	A machine that learns from data, thinks about it, solves problems and takes decisions — the way a person would.

A person: **Observe → Learn → Decide** ≡ A machine: **Data → Model → Prediction**

Today's AI is **Narrow AI**. It is very good at **one** job only. A program that plays chess beautifully cannot fill your tax form. There is no machine yet that can do everything a human can.

Old Programming vs Artificial Intelligence

Old (traditional) programming	Artificial Intelligence
We give the computer the rules . It gives us the answer.	We give the computer the answers and data . It finds the rules itself.
Same input always gives the same output	Gets better as it sees more data
Follows fixed instructions	Learns from examples
Cannot improve on its own	Improves continuously

Old programming: we give rules, it gives answers.

AI: we give answers, it finds the rules.

“Think of teaching a child what a cat is. You do not give a rule book about whiskers and tails. You show pictures and say ‘cat, cat, cat’. The child finds the rule alone. That is exactly how AI learns.”

MATHEMATICS + ALGORITHMS + DATA = ARTIFICIAL INTELLIGENCE

Programming is only the tool we type with. **Mathematics is the real language of AI.** To find patterns in data, a machine needs mathematical models, continuous calculation and careful optimization. Nothing else.

M.Sc. Mathematics background is a **big advantage**, not a disadvantage. Many people enter AI from outside. These students are entering from the foundation. AI cannot exist without mathematics.

The Six Branches You Need

Branch	What it does in AI
Logic	Helps the machine reason and draw conclusions
Probability	Handles uncertainty — “how likely is this?”
Linear Algebra	Stores data as vectors and matrices (the base of deep learning)
Statistics	Studies data and finds patterns in it
Optimization	Finds the best answer by reducing error
Graph Theory	Handles networks, connections and routes

The Mathematics Inside the Apps They Use

Area	App	Mathematics behind it
Navigation	Google Maps	Graph Theory + Optimization
Writing text	ChatGPT	Linear Algebra + Probability + Optimization
Seeing images	Face Recognition	Linear Algebra + Matrix operations
Suggestions	Netflix / Amazon	Probability + Statistics + Optimization
Safety	Spam filtering	Probability + Statistics

Ask every student to name **one** mathematics topic they have already studied, and say which app in the table above it could be working inside. Accept even partly correct answers. The aim is connection, not marks.

Keeping AI Safe: Cryptography and Cyber Security

“Mathematics makes AI possible. Now let us see how AI keeps us safe.”
Cryptography protects the data. AI studies the data. Two different jobs.

Point	Cryptography	Artificial Intelligence
Main job	Keeps information secret and safe	Learns from information
Nature	Fixed mathematical security methods	Adapts and improves
Gives us	Locked, verified data	A prediction or a decision

without cryptography, AI would receive wrong or changed data and learn the wrong thing. Without AI, today's cyber attacks would be too clever for us to catch. The two do not compete — they help each other.

Where AI Protects Us

Spam Finds harmful emails by recognising patterns.	Fraud Catches strange bank or UPI transactions at once.	Intrusion Notices suspicious activity on a network.
Malware Sorts harmful software by its behaviour.	Phishing Spots fake websites and fake emails.	Signatures Notices unusual signing behaviour.

ABOUT DIGITAL SIGNATURES

AI does not replace a digital signature. The signature still proves the document. AI simply watches the behaviour around it — who signed, from where, at what time — to check that nobody stole the key.

Looking Ahead: Quantum Computers

In simple words	
The problem	Quantum computers may one day break the security methods we use today.
The answer	Post-Quantum Cryptography (PQC) — new methods built on harder mathematics, such as lattices, that quantum computers cannot break.
Where AI helps	AI helps researchers choose good settings, test new attacks and find mistakes in the code.

FOR STUDENTS INTERESTED IN RESEARCH

Federated Learning Explainable AI Privacy-Preserving Machine Learning AI for Post-Quantum Cryptography
Mention these only briefly today. They are project and dissertation areas inside this M.Sc. programme, not exam topics for Unit I.

Putting It All Together

Step 1 · Raw Data The fuel	Step 2 · Cryptography The lock that protects it	Step 3 · Safe Data Data we can trust	Step 4 · Mathematics The engine that processes it	Step 5 · AI The decision that comes out
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“AI is not magic. It is safe data, passed through mathematics, to produce a useful decision. That is the whole story. Everything we study this semester sits somewhere in these five steps.”

Six Things to Remember from Today

- AI is already inside your ordinary day.
- Intelligence is a circle: observe, learn, reason, decide, adapt.
- Old programming gives rules. AI finds the rules.
- Mathematics + Algorithms + Data = AI.
- Cryptography protects data; AI learns from it.
- Today's AI is narrow — good at one job only.

HOMEWORK

1. **Report** — “AI in My Daily Life”: list 20 AI apps you use and write one line on what each one does.
2. **Chart** — “Mathematics Behind AI”: write one AI application for each of the six branches.

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